

Rehan Ashfaq Syed

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Data engineer with ML exposure building production-grade pipelines, feature stores, and data platforms on dbt, Databricks, Spark, Airflow, and GCP/AWS. Five years shipping large-scale ETL/ELT systems, orchestration workflows, and ML infrastructure across Bell, Conservice, and Kingscote – with hands-on experience supporting NLP and predictive modeling in production.

EXPERIENCE

Bell

Data Scientist

Aug 2022 – Present

- Built **Apache Flink + Redis** live streaming pipelines processing high-throughput CSP network telemetry, detecting disconnected wires in real time and generating **\$750K** annual savings.
- Engineered **Hadoop-to-GCP** data migration pipelines enriching ML training tables, improving downstream model precision 34%.
- Built **GCP** geospatial ETL pipelines using Python and Google Maps APIs, lifting customer address coordinate accuracy from 60% to 85%.
- Applied time-series feature engineering on high-frequency network telemetry to surface failure lead-lag relationships, cutting MTTD 25%.
- Built network disruption ML models: **TensorFlow** feedforward (87% accuracy) and **CNN-LSTM** time-series, cutting unnecessary truck rolls by 17%.
- Ran **Hadoop** network infrastructure analysis identifying CSP capacity gaps, securing **\$3.5M** for remediation.
- Productionized Python/Java data-pull jobs with DevOps, co-authoring **GitLab CI/CD** pipelines and Dockerfiles and tuning **OpenShift** CPU/memory allocations.
- Owned analytics for the Virtual Repair RL diagnostic system, building FMO vs CMO comparison frameworks quantifying model lift and regressions against current technician workflows.
- Built an **LLM agent** querying internal APIs and SQL DBs for recurring Outage Notification Program ad-hoc requests, saving 4 analyst-hours weekly.
- Built live **GCP Pub/Sub** streaming ingestion joined to Oracle, loading into **SQL Server** triggers for at-risk B2B ticket monitoring, cutting SLA breach times 80%.

Tools: Python, SQL, Apache Flink, Redis, GCP Pub/Sub, Hadoop, TensorFlow, GitLab CI/CD, Docker, OpenShift, GCP, Oracle, SQL Server, LLM Agents

Conservice

Data Scientist

Jan 2022 – Aug 2022

- Built production **Databricks (PySpark)** transformations on **Delta Lake** tables, unifying raw utility feeds into analytics-ready datasets for forecasting and anomaly models.
- Developed modular **dbt** models over **AWS Redshift** and Databricks (staging / intermediate / mart layers), improving lineage transparency and cutting analyst onboarding 40%.
- Implemented **dbt** tests and data quality monitoring (schema, uniqueness, not-null, freshness), catching upstream drift before downstream models broke.
- Orchestrated ML data pipelines in **Airflow and Dagster**, scheduling feature engineering jobs and maintaining reusable feature-store style datasets.
- Deployed ML pipelines with engineering using **Docker, Kubernetes, and CI/CD** for scalable model serving and reproducible releases.
- Applied **K-Means** clustering on 50K+ Redshift residential properties to segment by geography and consumption behavior, surfacing high-risk usage outliers.
- Built **XGBoost** and Linear Regression models forecasting monthly utility expenses across residential portfolios, improving property-manager budgeting accuracy.
- Ran time-series analysis on smart-meter data to detect seasonal and over-usage trends informing operational optimization.

Tools: Python, SQL, Databricks, PySpark, Delta Lake, dbt, AWS Redshift, Airflow, Dagster, Docker, Kubernetes, CI/CD, XG-Boost, Scikit-learn

Kingscote Informatics

Data Analyst Jan 2021 – Oct 2021

- Built modular **dbt + Snowflake** pipelines standardizing raw IoT sensor data into analytics-ready datasets, cutting manual prep time 40%.
- Authored reusable **dbt** macros and tests (uniqueness, not-null, accepted-value) across staging and mart models, hardening reporting reliability.
- Documented **dbt** model contracts and lineage in **Git**, giving consumers a single source of truth for sensor data definitions.
- Built **Airflow** orchestration workflows scheduling sensor telemetry and log ingestion ETL jobs, improving data availability for analytics teams.
- Engineered reusable feature pipelines for predictive maintenance, building curated feature-store style tables that improved training efficiency and reproducibility.
- Analyzed 86K+ IoT sensor logs in **SQL and R** to identify noise and signal degradation, improving data quality for downstream analytics.
- Built **Streamlit** prototypes exposing sensor health metrics to operations teams, accelerating triage on field-deployed hardware.

Tools: dbt, Snowflake, Airflow, SQL, R, Streamlit, Git, ETL Pipelines, Feature Engineering, Feature Store

PROJECTS

- **Open Intent Classifier** – End-to-end NLP pipeline (**BERT + RAG**) with ingestion, embeddings, and retrieval, achieving 30% higher accuracy over a graph-based baseline.
- **Sentiment Classification** – **LSTM + Self-Attention** model on IMDb reviews achieving 97.4% accuracy, deployed via Streamlit.
- **RFID Embedded Speech Synthesizer** – Embedded system reading RFID tag IDs aloud via a speech-synthesis module, integrating microcontroller firmware with the RFID reader.

EDUCATION

Master of Engineering – Electrical & Computer Engineering	<i>Dec 2025</i>
Toronto Metropolitan University	
Bachelor of Engineering – Electronics and Communication Engineering	<i>Sep 2020</i>
Osmania University	

SKILLS

Data Engineering	dbt, Databricks, PySpark, Delta Lake, Snowflake, Apache Flink, Airflow, Dagster, ETL/ELT Pipelines, Feature Store
Cloud & Big Data	GCP, AWS, BigQuery, Redshift, Spark, Hadoop, Kafka, GCP Pub/Sub
DevOps & MLOps	Docker, Kubernetes, OpenShift, GitLab CI/CD, Git, MLOps, ML Pipelines
Machine Learning & NLP	TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, BERT, NLP, LLM Agents, RAG
Databases	SQL, NoSQL, MongoDB, Cassandra, Redis, Elasticsearch, Oracle, SQL Server
Analytics & BI	Looker (LookML), Power BI, Grafana, Plotly Dash, Streamlit
Core	Python, SQL, R, Java, Machine Learning

CERTIFICATIONS & AWARDS

• Merit Distinction Award – Q1 2023 & Q3 2024	Bell
• Python for Data Science	IBM
• Machine Learning with Python	IBM
• Database Design and Programming	Oracle
• Google Cloud Certification for GCP	Google